

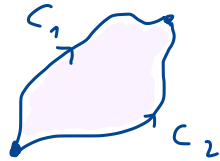
Supplemental lecture

Complex series, analyticity

Cauchy's integral representation

Contour deformation:

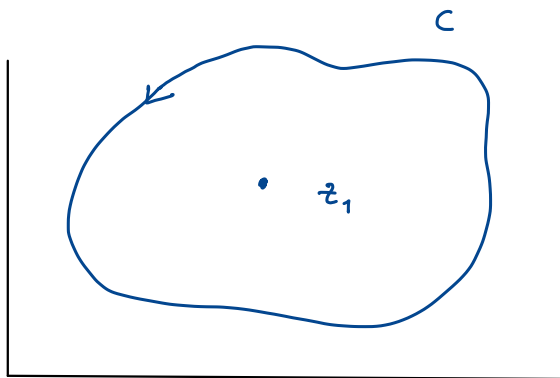
Cauchy:



$$\int_{C_1} dz f(z) = \int_{C_2} dz f(z)$$

if $f(z)$ differentiable in the pink region.

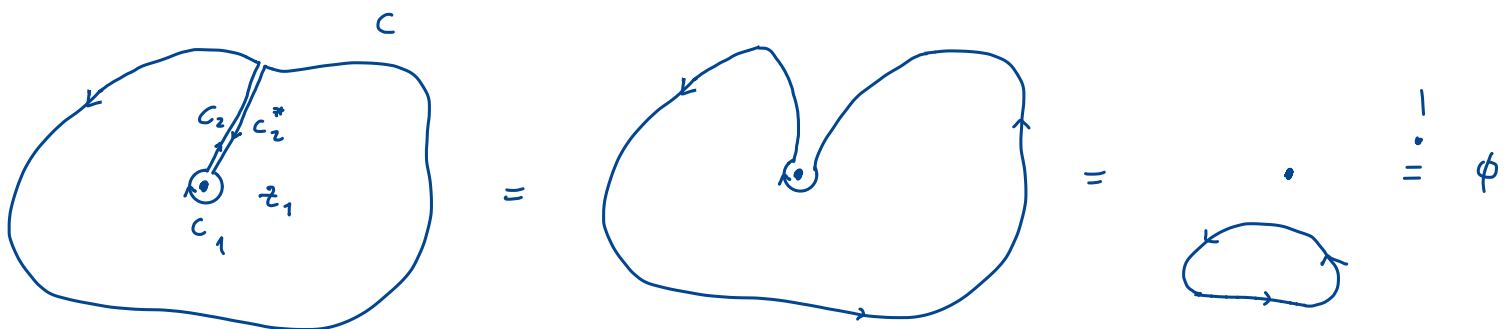
A simple integral:



$$\oint_C \frac{1}{z - z_1} dz = 2\pi i$$

Proof:

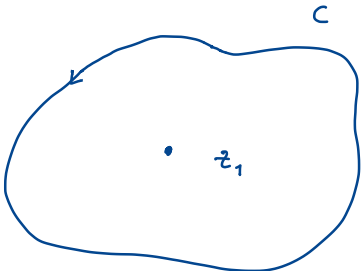
consider integral



$$\Rightarrow \int_{C_1} f(z) dz + \int_C dz f(z) = 0 \quad \text{since} \quad \int_{C_2} \dots = - \int_{C_2^*}$$

$\uparrow$
 $f(z) = \frac{1}{z - z_1}$ now

$$\text{or: } \int_C dz f(z) = \int_{C_1^*} f(z) dz$$

$$I = \oint_C = \oint_{C_1^*}^{z_1} = \oint_{C_1^*} dz \frac{1}{z - z_1}$$


parametrization:

$$z \equiv z_1 + \rho \cdot e^{i\varphi}$$

$$dz = i d\varphi \cdot \rho \cdot e^{i\varphi}$$

$$\Rightarrow I = \int_0^{2\pi} d\varphi : \cancel{\rho e^{i\varphi}} \frac{1}{\cancel{\rho e^{i\varphi}}} = 2\pi i \quad \checkmark$$

generalization:

$$\oint_C (z - z_1)^m = \begin{cases} 2\pi i & m = -1 \\ \emptyset & \text{otherwise} \end{cases}$$

Thm. 1, Cauchy's integral representation
 $f(z)$ differentiable on a simple domain D ,
 $C \subset D$ simple closed curve around $z_1 \in D$

$\Rightarrow$

$$\oint_C \frac{f(z)}{z - z_1} dz = 2\pi i \cdot f(z_1)$$

For the proof we use the inequality:

LM inequality:

$$\left| \int_C f(z) dz \right| \leq L \cdot \pi$$

length of C $\max_{z \in C} |f(z)|$

proof relies on triangle inequality:

$$z_1, z_2 \in \mathbb{C} \Rightarrow |z_1 + z_2| \leq |z_1| + |z_2|$$

$$\int_C f(z) dz = \lim_{n \rightarrow \infty} \sum_{i=1}^n \Delta z_i f(z_i)$$

$$\begin{aligned} \left| \sum_{i=1}^n \Delta z_i f(z_i) \right| &\leq \sum_{i=1}^n |\Delta z_i f(z_i)| = \\ &= \sum_{i=1}^n |\Delta z_i| |f(z_i)| \leq \max_i |f(z_i)| \cdot \sum_{i=1}^n |\Delta z_i| \leq \\ &\leq \max_{z \in C} |f(z)| \cdot \sum_{i=1}^n |\Delta z_i| \end{aligned}$$

$$\Rightarrow \left| \int_C f(z) dz \right| = \lim_{n \rightarrow \infty} \left| \sum_{i=1}^n \Delta z_i f(z_i) \right| \leq$$

$$\leq \underbrace{\left(\max_{z \in C} |f(z)| \right)}_{\pi} \cdot \underbrace{\lim_{n \rightarrow \infty} \sum_{i=1}^n |\Delta z_i|}_{L} = \pi \cdot L \quad \checkmark$$

proof of Thm 1:

$$\oint_C \frac{f(z)}{z - z_1} dz = \oint_{C_1^*} \frac{f(z)}{z - z_1} dz =$$

$\uparrow$
 differentiable
 in $D \setminus \{z_1\}$

$\uparrow$
 circle of small
 radius δ
 around z_1

$$= \underbrace{\oint_{C_1^*} \frac{f(z_1)}{z - z_1} dz}_{f(z_1) \cdot 2\pi i} + \underbrace{\oint_{C_1^*} \frac{f(z) - f(z_1)}{z - z_1} dz}_R$$

We now show that $R = 0$.

$f(z)$ differentiable $\Rightarrow$ continuous $\Rightarrow$

$\forall \epsilon \in \mathbb{R} \exists \delta$ such that $|f(z) - f(z_1)| \leq \epsilon$
 for any $|z - z_1| \leq \delta$

then:

$$|R| \leq L \cdot \pi = (2\pi \cdot \delta) \cdot \max_{z \in C_1^*} \frac{|f(z) - f(z_1)|}{|z - z_1|} =$$

$$= 2\pi \max_{z \in C_1^*} |f(z) - f(z_1)| \leq 2\pi \cdot \epsilon$$

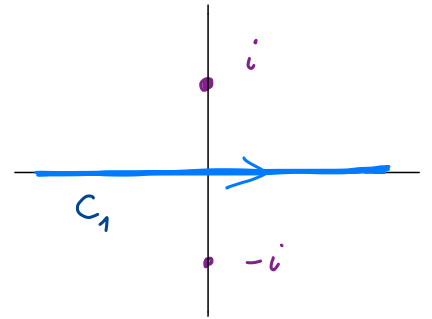
$$|R| \leq 2\pi \cdot \epsilon \quad \text{for any } \epsilon \Rightarrow R = 0 \quad ! \quad \checkmark$$

Application:

We can use this thm. to evaluate integrals.

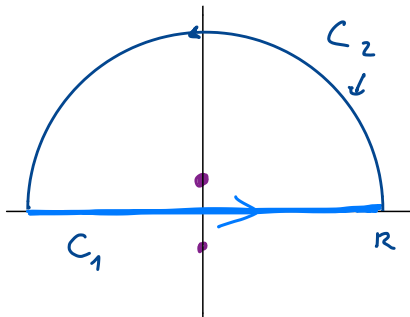
Consider, e.g.,
$$I = \int_{-\infty}^{\infty} dx \frac{1}{x^2+1} = ?$$

$$\int_{-\infty}^{\infty} dz \frac{1}{z^2+1} = \int_{C_1} dz \frac{1}{z-i} \frac{1}{z+i}$$



Close contour:

as $R \rightarrow \infty$



$$\int_{C_2} \frac{1}{z^2+1} dz \leq \pi \cdot R \max \frac{1}{|z^2+1|} \leq$$

$$\leq \pi \cdot R \max \frac{1}{|z^2|-1} = \frac{\pi \cdot R}{R^2-1} \rightarrow 0$$

$$\Rightarrow I = \oint_{C_1+C_2} \frac{1}{z-i} \frac{1}{z+i} dz = 2\pi i \cdot \underset{\substack{\uparrow \\ \frac{1}{z-z_1}}}{f(z)} \underset{\substack{\uparrow \\ f(z_1) \\ z_1=i}}{f(z=i)} = \underset{\substack{\uparrow \\ \frac{1}{z \cdot i}}}{2\pi i} = \pi \quad \checkmark$$

Home work:

$$\int_{-\infty}^{\infty} \frac{1}{x^4+1} dx = ?$$

Consequence:

Thm. 2. $f(z)$ differentiable over D simply connected $\Rightarrow$ infinitely differentiable (!) and

$$f^{(m)}(z) = \frac{m!}{2\pi i} \oint_C dz^* \frac{1}{(z^* - z)^{m+1}} f(z^*)$$

with $C \subset D$ any simple curve around z .

Proof rigorous proof uses L.M inequality ...
one can, however, make it plausible:

$$\text{Thm 1: } f(z) = \frac{1}{2\pi i} \oint_C \frac{1}{z^* - z} f(z^*) dz^* = I(z)$$

$\text{Re } f(z)$ and $\text{Im } f(z)$ are just line integrals of non-singular functions for $z^* \in C$,
and $z = x + iy$, i.e., x and y are just parameters!

$$\Rightarrow \frac{df}{dz} = \frac{dI}{dz} = \frac{1}{2\pi i} \oint_C \underbrace{\frac{\partial}{\partial z} \left(\frac{1}{z^* - z} f(z^*) \right)}_{\frac{1}{(z^* - z)^2} f(z^*)} dz^*$$

similarly: $f^{(m)}(z) = \frac{d^m f}{dz^m} =$

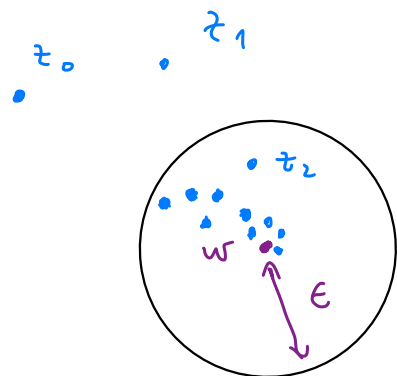
$$= \frac{1}{2\pi i} \oint_C \frac{\partial^m}{\partial z^m} \left(\frac{1}{z^* - z} f(z^*) \right) dz^* = \frac{1}{2\pi i} \oint_C \frac{m!}{(z^* - z)^{m+1}} f(z^*) dz^*$$

✓

Complex power series

Series $\{z_n\} = \{z_0, z_1, \dots\}$

Def: z_i converges to $w \in \mathbb{C}$ if
 $\forall \epsilon \exists N$ that for $\forall n \geq N$ $|z_n - w| < \epsilon$



Sum: $S_n \equiv \sum_{k=0}^n z_k$

converges if $\exists S = \lim_{n \rightarrow \infty} S_n$

Def.: Power series around z_0 .

- $\{a_n\}$ complex series
- $f_n(z) \equiv \sum_{m=0}^n a_m \cdot (z - z_0)^m$

$$f(z) \equiv \lim_{n \rightarrow \infty} f_n(z) \quad \text{if converges}$$

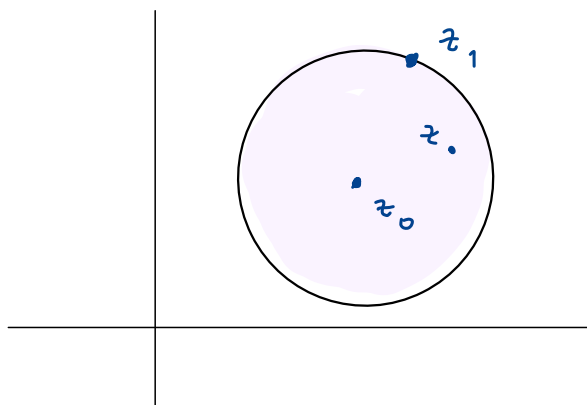
or

$$f(z) = \sum_{m=0}^{\infty} a_m (z - z_0)^m$$

convergence radius

• if $\exists \sum_{n=0}^{\infty} a_n (z_1 - z_0)^n$ then

$\exists \sum_{n=0}^{\infty} a_n \cdot (z - z_0)^n$ for any $|z - z_0| < |z_1 - z_0|!$



• convergence radius:

$$R = \lim \left| \frac{a_n}{a_{n+1}} \right|$$

statement: $f(z) = \sum_{n=0}^{\infty} a_n \cdot (z - z_0)^n$

converges for any z , $|z - z_0| < R$,
but does not converge for $|z - z_0| > R$!

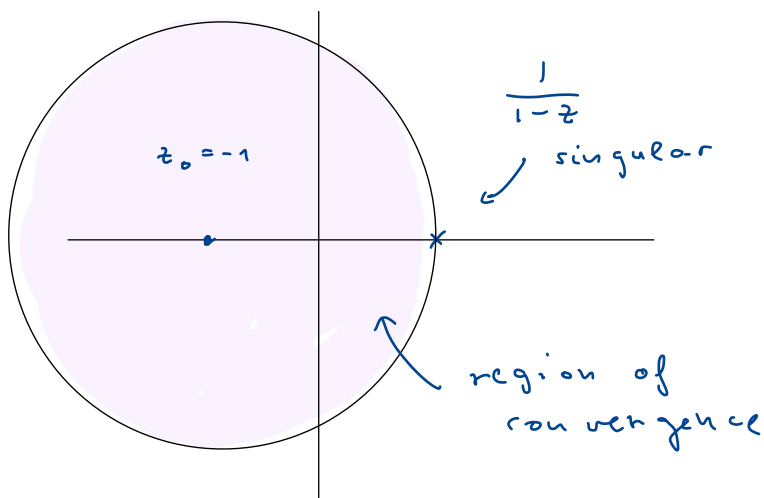
examples:

• $\frac{1}{1-z} = 1 + z + z^2 + \dots$

$a_n = 1 \Rightarrow R = 1$ for $z_0 = 0$

• $\frac{1}{1-z} = \frac{1}{2-(z+1)} = \frac{1}{2} + \frac{1}{2^2} (z+1) + \frac{1}{2^3} (z+1)^2 + \dots$

$z_0 = -1 \Rightarrow R = 2$!



$$\exp(z) = \sum_{m=0}^{\infty} \frac{1}{m!} z^m$$

$$\frac{a_m}{a_{m+1}} = \frac{(m+1)!}{m!} = m+1 \xrightarrow[m]{\uparrow} \infty \Rightarrow R = \infty$$

Thm: If $f(z) = \sum_{m=0}^{\infty} a_m \cdot (z - z_0)^m$

converges for $|z - z_0| < R \neq 0 \Rightarrow$

$\exists f'(z)$ for $|z - z_0| < R$ and

$$f'(z) = \sum_{m=0}^{\infty} b_m (z - z_0)^m$$

with $b_m = (m+1) \cdot a_{m+1}$ (proof easy)

Corollary: $f(z_0) = a_0$, $f'(z_0) = a_1$, ...,

$$\dots f^{(m)}(z_0) = m! \cdot a_m$$

$$\Rightarrow a_m = \frac{1}{m!} f^{(m)}(z_0) = \frac{1}{2\pi i} \oint_C \frac{1}{(z - z_0)^{m+1}} f(z)$$

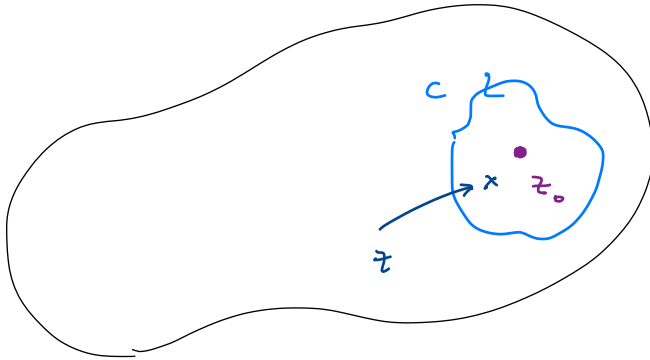
Thm: $f(z)$ differentiable over $D \Rightarrow$

Taylor series

$$f(z) \equiv \sum_{m=0}^{\infty} \frac{1}{m!} f^{(m)}\left(\frac{z}{z_0}\right) \cdot (z - z_0)^m$$

converges in a finite neighborhood of

ANY $z_0 \in D$.



Proof (sketch)

$$\text{Cauchy: } f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z^*)}{z^* - z_0} dz^*$$

idea: use same contour around $z \approx z_0$.

$$(2\pi \cdot i) \cdot f(z) = \oint dz^* \frac{f(z^*)}{z^* - z}$$

$$\frac{1}{z^* - z} = \frac{1}{z^* - z_0 - (z - z_0)} = \frac{1}{z^* - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{z^* - z_0}} =$$

$$= \frac{1}{z^* - z_0} \left(1 + \frac{z - z_0}{z^* - z_0} + \frac{(z - z_0)^2}{(z^* - z_0)^2} + \dots \right)$$

$$= \sum_{m=0}^{\infty} \frac{1}{(z^* - z_0)^{m+1}} \cdot (z - z_0)^m$$

$$\Rightarrow \text{2nd: } f(z) = \oint_C f(z^*) \cdot \sum_{m=0}^{\infty} \frac{1}{(z^* - z_0)^{m+1}} \cdot (z - z_0)^m dz^*$$

$$= \sum_{m=0}^{\infty} (z - z_0)^m \cdot \underbrace{\oint_C \frac{1}{(z^* - z_0)^{m+1}} \cdot f(z^*) dz^*}_{\frac{1}{m!} f^{(m)}(z_0)} \Rightarrow$$

$$\Rightarrow f(z) = \sum_{m=0}^{\infty} \frac{1}{m!} f^{(m)}(z_0) \cdot (z - z_0)^m$$

↑
series converges in a neighborhood!
(has to be shown...)

Laurent series

Series concept can be extended to negative powers:

$$f(z) = \sum_{m=-\infty}^{\infty} a_m (z - z_0)^m \quad \text{Laurent series}$$

- $m < 0$ part singular at z_0
- converges in a ring around z_0
- $a_m = \frac{1}{2\pi i} \oint_C dz^* \frac{f(z^*)}{(z^* - z_0)^{m+1}} \quad \underline{m \in \mathbb{Z}!}$

example:

$$e^{1/z} = 1 + \frac{1}{z} + \frac{1}{2} \frac{1}{z^2} + \dots = \sum_{n=0}^{\infty} \frac{1}{n!} \frac{1}{z^n}$$

- has essential singularity at $z=0$
- analytical at any $z \neq 0$

Residue theorem:

$f(z)$ differentiable but at z_0

$$\Rightarrow \oint_C f(z) dz = 2\pi i \operatorname{Res}_{z=z_0}(f(z)) = 2\pi i \cdot a_{-1}$$

example: $f(z) = \frac{1}{z^2 - 2z^3} = \frac{1}{z^2} \frac{1}{1 - 2z} =$

$$= \frac{1}{z^2} (1 + 2z + 4z^2 + \dots) = \underbrace{\frac{1}{z^2} + \frac{2}{z} + 4 + \dots}_{\text{singular part}} \quad \text{residue!}$$